Regional Labor Market in the United States

ECON10 — Economics of World Economy

Your Name

2026-04-05

Overview

This report examines **state-level unemployment rates** across the United States using monthly data from the Bureau of Labor Statistics (BLS). We walk through three steps:

1. **Data Collection** — where the data comes from and how it was obtained
2. **Data Cleaning** — how the raw data was reshaped and prepared for analysis
3. **Data Visualization** — a time trend chart and choropleth maps (static and interactive)

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1. Data Source

Item	Detail
Source	U.S. Bureau of Labor Statistics (BLS)
Program	Local Area Unemployment Statistics (LAUS)
File	`la.data.3.AllStatesS` — Seasonally Adjusted Monthly Unemployment Rates by State
URL	https://download.bls.gov/pub/time.series/la/
Downloaded	April 5, 2026
Coverage	All 50 U.S. states + D.C., January 1976 – present
Unit	Unemployment rate (percent, seasonally adjusted)

> **Note:** Seasonal adjustment removes the effect of predictable seasonal patterns (e.g., holiday hiring, summer farm labor), making month-to-month comparisons more meaningful.

Exact Data Used In This Report

This analysis explicitly uses the following BLS file and filter:

- **Primary dataset file:** `la.data.3.AllStatesS`
- **Direct URL:** https://download.bls.gov/pub/time.series/la/la.data.3.AllStatesS
- **Local project file used:** `data/la.data.3.AllStatesS.txt`
- **Series filter applied:** `series_id` ending in `003` only
- **Meaning of measure code:** `03 = unemployment rate` (from BLS LAUS measure mapping)
- **Documentation URL:** https://download.bls.gov/pub/time.series/la/la.txt
- **Measure mapping URL:** https://download.bls.gov/pub/time.series/la/la.measure

This report does **not** use the other measure codes in the same file (e.g., `004`, `005`, `006`, `007`, `008`).

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2. Data Loading

The raw file is a tab-separated text file. Each row represents one state's unemployment rate for one month.

```
# The raw file looks like this:
# series_id                    year  period  value  footnote_codes
# LASST010000000000003          1976  M01     6.7
#
# The "series_id" encodes which state:
# - "LASST" = BLS prefix for state-level series
# - Characters 6-7 (0-indexed 5-6) = 2-digit state FIPS code (e.g., "01" = Alabama)
# - The last digit "3" = unemployment rate measure
```

```
Using series suffix: ['003']
Total rows: 31,200
```

	series_id	year	period	value	footnote_codes
0	LASST010000000000003	1976	M01	6.7	NaN
1	LASST010000000000003	1976	M02	6.6	NaN
2	LASST010000000000003	1976	M03	6.6	NaN
3	LASST010000000000003	1976	M04	6.5	NaN

	series_id	year	period	value	footnote_codes
4	LASST010000000000003	1976	M05	6.4	NaN

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3. Data Cleaning

Step 1 — Extract State FIPS Code from the Series ID

The series ID encodes which state it refers to. We extract characters at positions 5–6 to get the two-digit FIPS code.

```
Unique state FIPS codes: 52
<StringArray>
['01', '02', '04', '05', '06', '08', '09', '10', '11', '12', '13', '15', '16',
 '17', '18', '19', '20', '21', '22', '23', '24', '25', '26', '27', '28', '29',
 '30', '31', '32', '33', '34', '35', '36', '37', '38', '39', '40', '41', '42',
 '44', '45', '46', '47', '48', '49', '50', '51', '53', '54', '55', '56', '72']
Length: 52, dtype: str
```

Step 2 — Map FIPS Codes to State Names and Abbreviations

FIPS codes are numbers assigned to each state by the federal government. We create a lookup table to convert them to readable state names.

```
Rows after cleaning: 31,200
```

	series_id	state_fips	state	state_abbr	year	period	value
0	LASST010000000000003	01	Alabama	AL	1976	M01	6.7
1	LASST010000000000003	01	Alabama	AL	1976	M02	6.6
2	LASST010000000000003	01	Alabama	AL	1976	M03	6.6
3	LASST010000000000003	01	Alabama	AL	1976	M04	6.5
4	LASST010000000000003	01	Alabama	AL	1976	M05	6.4
5	LASST010000000000003	01	Alabama	AL	1976	M06	6.5
6	LASST010000000000003	01	Alabama	AL	1976	M07	6.5
7	LASST010000000000003	01	Alabama	AL	1976	M08	6.7

Step 3 — Parse Year and Month into a Date Column

The `period` column is formatted as `M01`, `M02`, …, `M12`. We convert this into a proper date.

```
Date range: January 1976 to December 2025
```

	state	state_abbr	state_fips	date	year	month	unemployment_rate
0	Alabama	AL	01	1976-01-01	1976	1	6.7
1	Alabama	AL	01	1976-02-01	1976	2	6.6
2	Alabama	AL	01	1976-03-01	1976	3	6.6
3	Alabama	AL	01	1976-04-01	1976	4	6.5
4	Alabama	AL	01	1976-05-01	1976	5	6.4
5	Alabama	AL	01	1976-06-01	1976	6	6.5
6	Alabama	AL	01	1976-07-01	1976	7	6.5
7	Alabama	AL	01	1976-08-01	1976	8	6.7
8	Alabama	AL	01	1976-09-01	1976	9	6.8
9	Alabama	AL	01	1976-10-01	1976	10	6.9

Cleaning Summary

Step	Action	Reason
Column trimming	Removed extra whitespace from column names and `series_id`	Raw BLS files have padded spaces
FIPS extraction	Extracted characters 5–6 from `series_id`	State identifier is embedded in the series code
State name mapping	Mapped 2-digit FIPS → state name and abbreviation	Raw data has no human-readable state labels
Date parsing	Combined `year` + `period` → proper date column	Enables time-series plotting
Column selection	Kept only relevant columns	Removes unnecessary metadata

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4. Visualization

4a — Time Trend: National Average Unemployment Rate

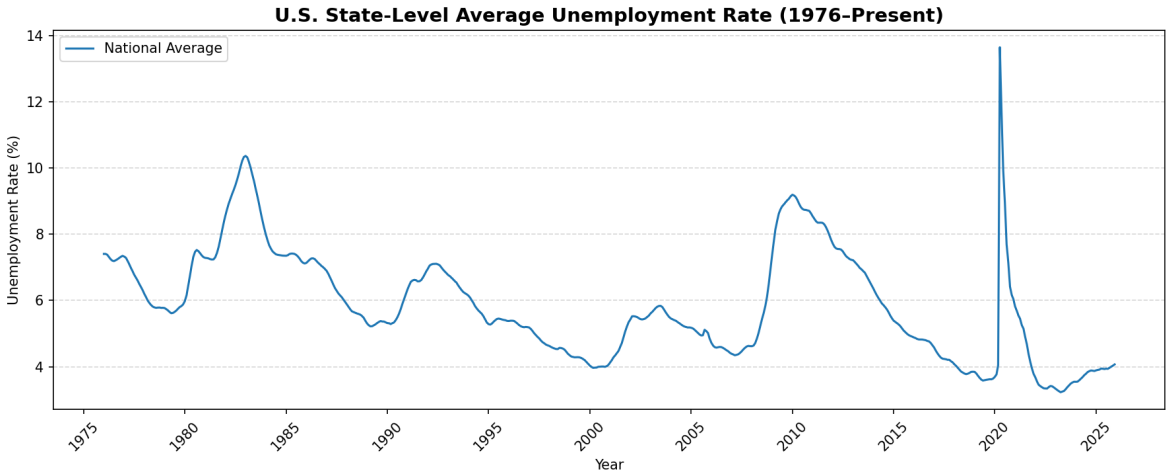

Figure 1: Figure 1. U.S. state-level average unemployment rate (monthly, seasonally adjusted).

4b — Time Trend: Selected States (Interactive)

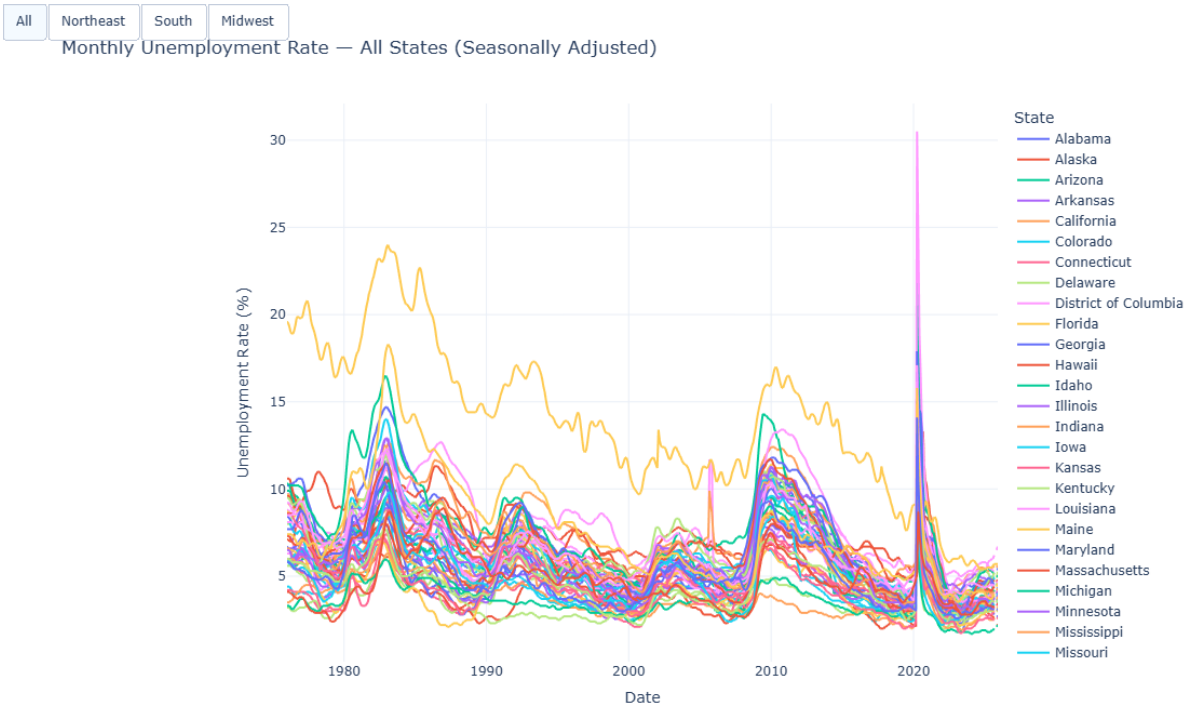

Figure 2: Figure 2. Monthly unemployment rate for all states. Use region filters (All/Northeast/South/Midwest) in the HTML version; this PDF shows a static snapshot.

4c — Static Map: Latest State Unemployment Rate

```
Mapping data for: December 2025
```

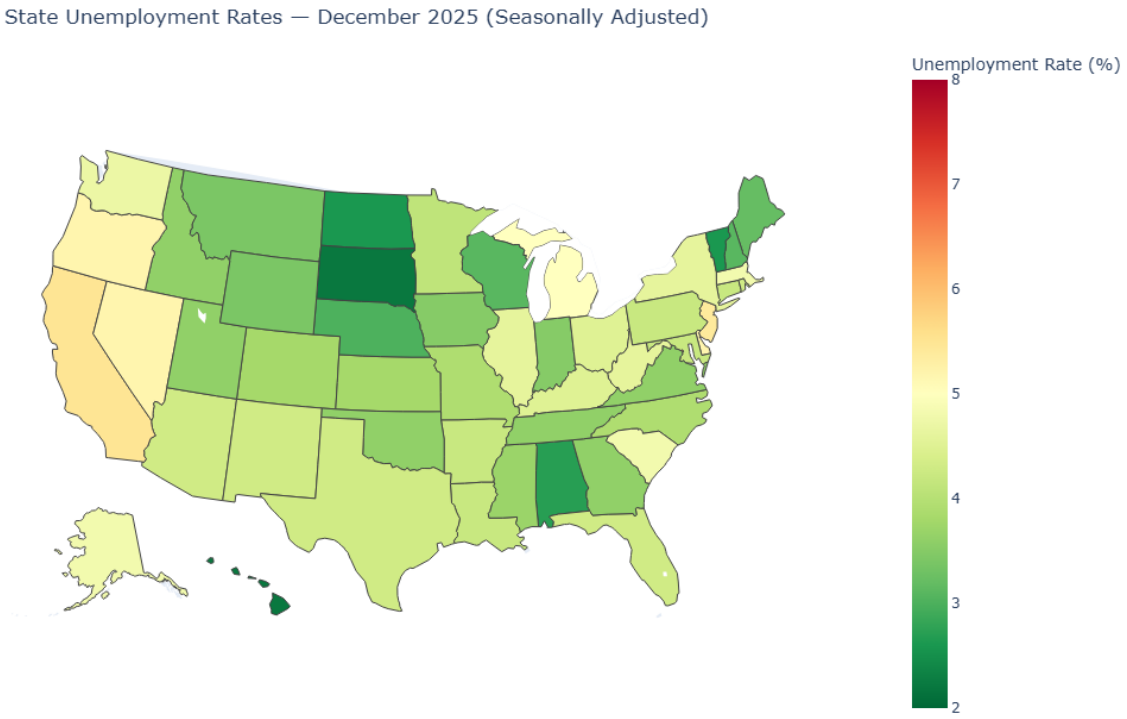

Figure 3: Figure 3. State unemployment rates in the latest complete month (seasonally adjusted).

4d — Static Map: Change from Oldest to Latest Complete Month

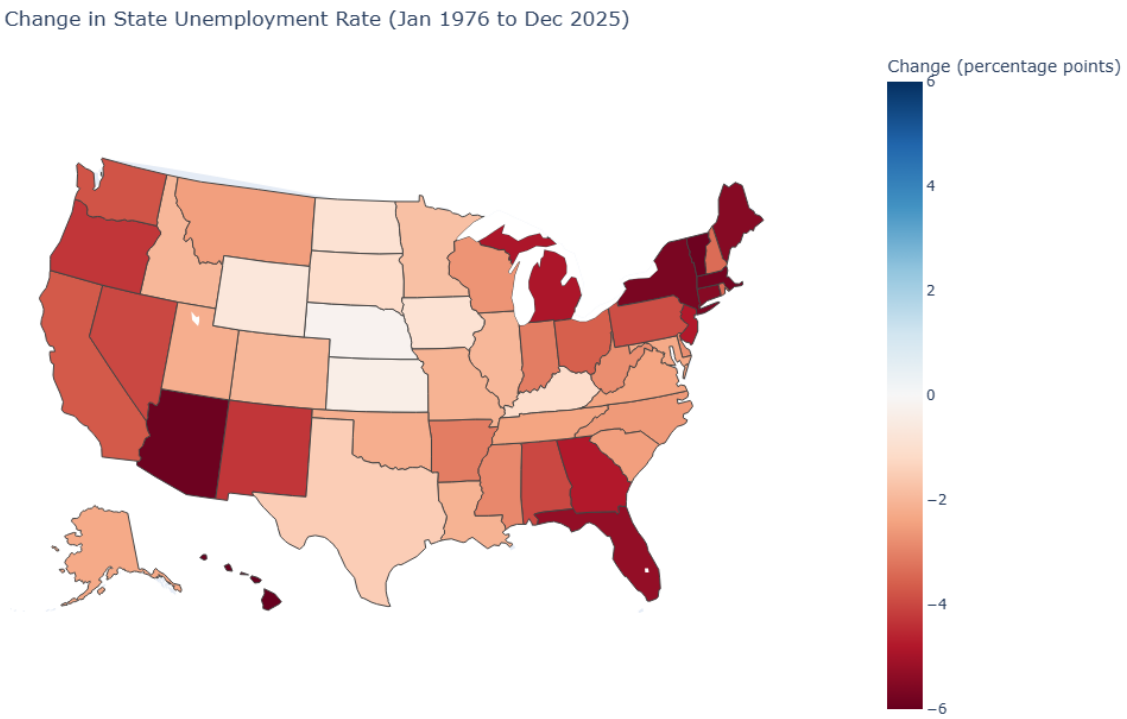

Figure 4: Figure 4. Change in state unemployment rates from the oldest complete month to the latest complete month (percentage points).

4e — Static Map: Recovery from COVID (2021 to 2024)

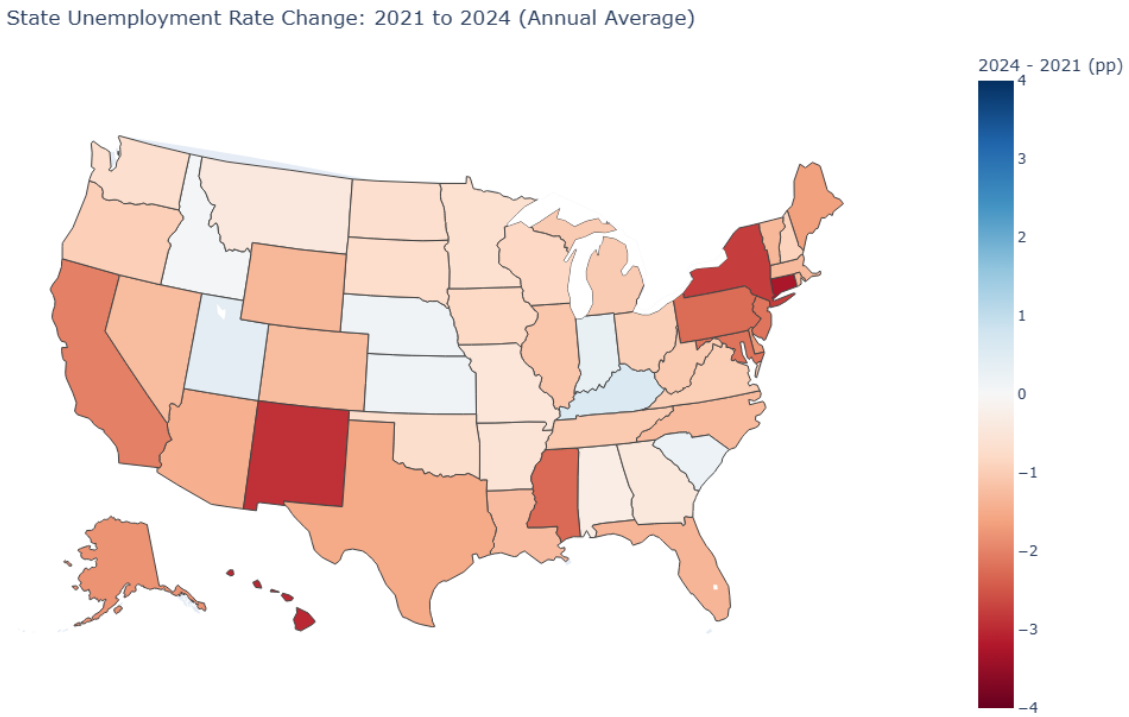

Figure 5: Figure 5. Change in annual-average unemployment rates from 2021 to 2024 (percentage points), showing post-COVID labor market recovery patterns.

4f — Static Map: Relative to Massachusetts

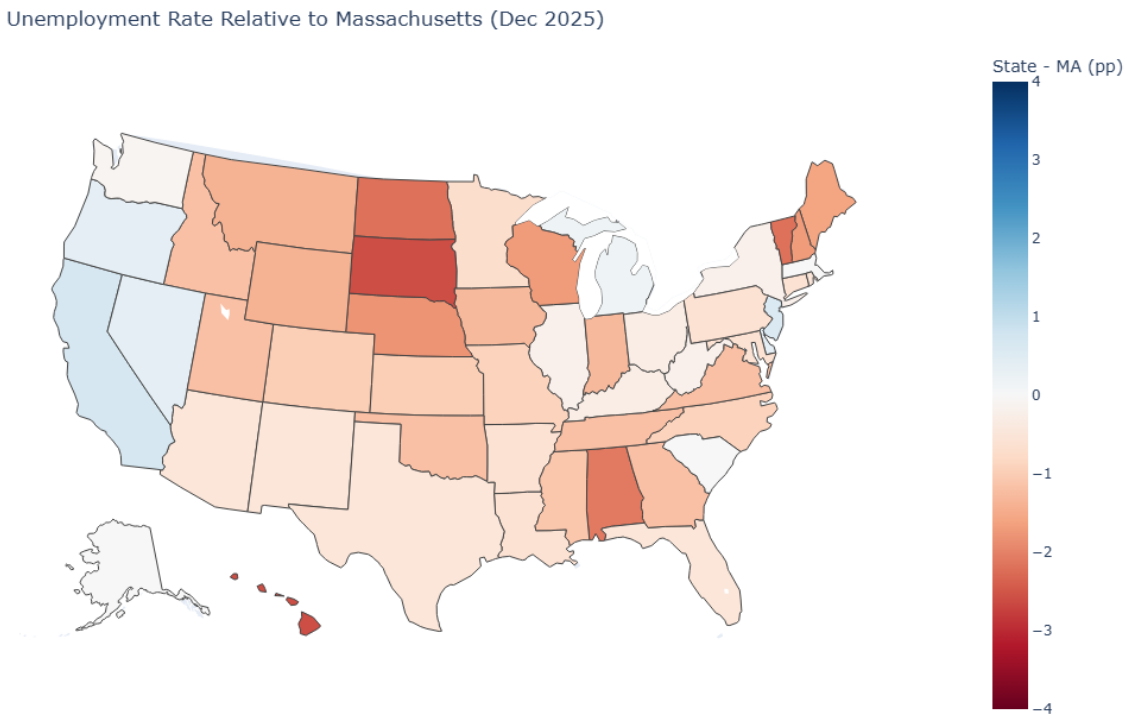

Figure 6: Figure 6. State unemployment rates relative to Massachusetts in the latest complete month (state rate minus MA rate, in percentage points).

4g — Static Map: Latest Annual Average (from annual panel)

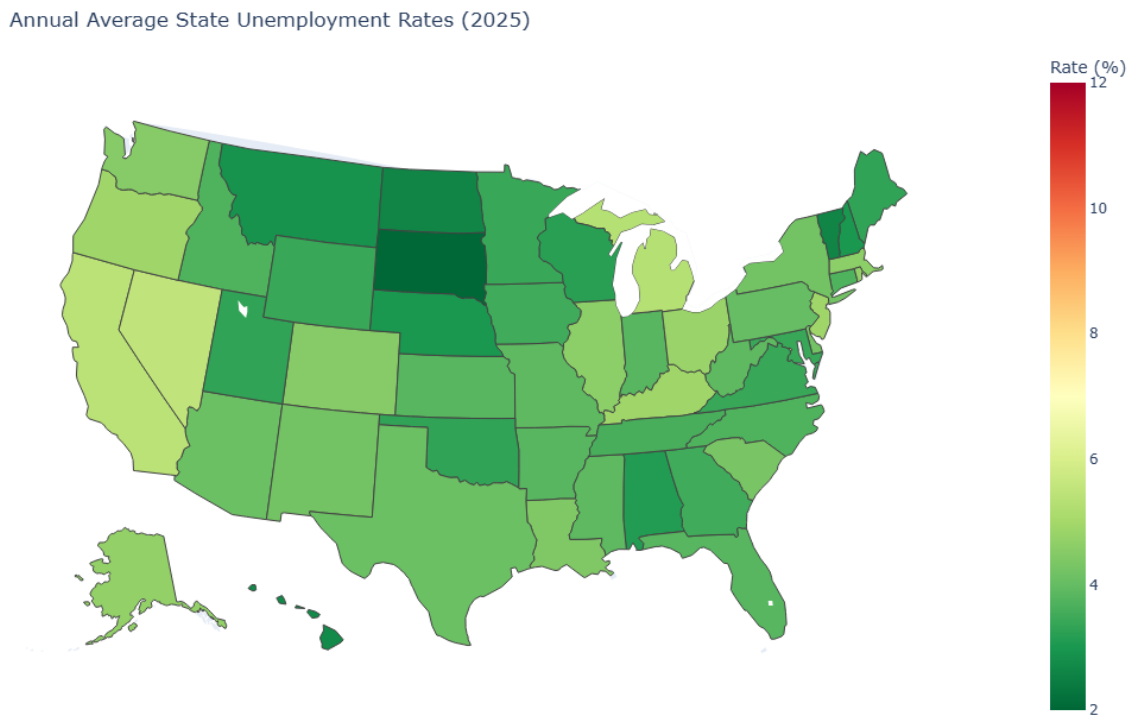

Figure 7: Figure 7. Annual-average unemployment rates by state in the latest available year.

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5. Key Observations

Where To Add Your Findings (Fill-In Section)

Use this section to summarize the main empirical findings from Figures 1–7.

Finding 1: National trend (Figure 1)

[Write 2-4 sentences: overall pattern, major turning points, and recent level.]

Finding 2: Regional/state patterns (Figures 2 and 3)

[Write 2-4 sentences: which regions/states are higher or lower, and whether dispersion across states is large or small.]

Finding 3: Long-run change and COVID recovery (Figures 4 and 5)

[Write 2-4 sentences: which states improved more/less, and how 2021 to 2024 compares across regions.]

Finding 4: Relative to Massachusetts (Figure 6)

[Write 2-4 sentences: which states are above MA, which are below MA, and what that implies for local comparison.]

Finding 5: Latest annual snapshot (Figure 7)

[Write 2-4 sentences: summarize the latest cross-state landscape in one paragraph.]

Short Discussion (Interpretation)

Add a brief discussion (about 1 short paragraph each):

1. **Economic interpretation:** [What labor market mechanisms might explain these patterns?]
2. **Policy relevance:** [What does this imply for unemployment policy or wage policy discussions?]
3. **Limitations:** [What this analysis does not capture (e.g., causality, demographics, industry composition).]
4. **Next step:** [One additional dataset or model you would add in a follow-up analysis.]

Slide-Ready Summary (3 bullets)

[Bullet 1: Main takeaway]

[Bullet 2: Evidence from maps/charts]

[Bullet 3: Caution or limitation]

Appendix: Data Citation

U.S. Bureau of Labor Statistics. (2026). *Local Area Unemployment Statistics (LAUS): Seasonally Adjusted State Unemployment Rates* [Data file]. Retrieved April 5, 2026, from https://download.bls.gov/pub/time.series/la/la.data.3.AllStatesS

U.S. Bureau of Labor Statistics. (2026). *LAUS Documentation* [Technical documentation]. Retrieved April 5, 2026, from https://download.bls.gov/pub/time.series/la/la.txt

U.S. Bureau of Labor Statistics. (2026). *LAUS Measure Mapping (la.measure)* [Codebook]. Retrieved April 5, 2026, from https://download.bls.gov/pub/time.series/la/la.measure